Face Recognition-Based Hotel Check-In & Check-Out System

A Python-based Face Recognition System designed to simplify and automate hotel guest check-in and check-out processes using real-time facial recognition technology.

The system uses **OpenCV** and the **face_recognition** library to identify registered individuals through a webcam, helping reduce manual identity verification and making the hotel registration experience faster and more convenient.

Key Features

- Real-time face detection using a webcam.
- Facial recognition of registered individuals.
- Automatic identification using stored facial encodings.
- Displays the recognized person's name on the screen.
- Identifies unregistered individuals as unknown persons.

Technologies Used

- Python
- OpenCV
- Face Recognition
- NumPy

Project Objective

To explore how AI-powered facial recognition can improve hotel operations by making guest identification easier, reducing manual check-in procedures, and creating a smoother check-in and check-out experience.

Future Enhancements

- Automated guest check-in and check-out.
- Guest database integration.
- Check-in and check-out timestamp recording.
- Automated billing and room allocation.
- Web-based hotel management dashboard.

Note: This is a prototype demonstrating facial recognition. Actual hotel deployment would require secure identity verification, guest consent, data protection, and integration with a hotel management system.